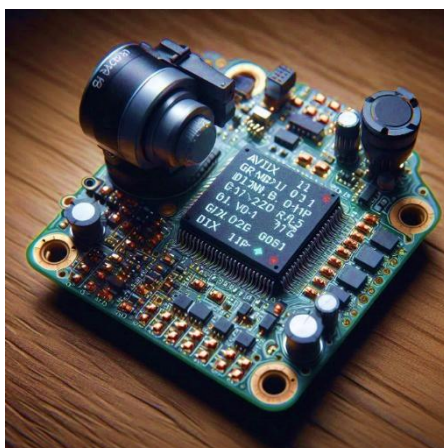




GIMBAL CONTROL MODULE

Interface Control Document V1.12



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AVIX TECHNOLOGY INC

No. 143, Sec. 3, Yongping Rd., Taiping Dist., Taichung City 411004, Taiwan (R.O.C.)

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V1.12	Update Section 1.1	Skye Tsai	2026.2.25
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Contents

Document Revision History	2
Contents	4
1. Gimbal Control Description	7
1.1. Communication Protocol Format	8
1.1.1. Protocol Packet Examples	9
1.2. Device ID Definition	10
1.3. Message Data Definition	11
1.3.1. Message Response	11
1.3.2. Follow Random Head	12
1.3.3. Control Gimbal Velocity	13
1.3.4. Transfer to Specify Angle	14
1.3.5. Camera Control	15
1.3.6. Camera Capture	17
1.3.7. Thermal Control	18
1.3.8. Distance Measurement	19
1.3.9. UAV Status Return	20

1.3.10. Get Control IP	22
1.3.11. Return / Change Control IP.....	23
1.3.12. Change Control IP Response.....	24
1.3.13. Gimbal Status Return	25
1.3.14. Gimbal Version Return.....	27
1.3.15. Get Gimbal Version	29
1.4. Checksum Calculation	30
1.4.1. C++ Example	30
1.4.2. Python Example	30
2. Camera Control Description.....	31
2.1. Message Data Definition.....	32
2.1.1. SetPTZ.....	32
2.1.2. SetThermal	34
2.1.3. GetAdjust.....	35
2.1.4. SetAdjust.....	37
2.1.5. GetNetwork.....	39
2.1.6. SetNetwork	41

2.1.7. GetMultimedia	42
2.1.8. GetImage	44
2.1.9. SetMultimedia	46
2.1.10. GetPlayBack	48
2.1.11. GetSRTLlist.....	50
2.1.12. GetTime	51
2.1.13. SetTime.....	53
2.1.14. SetAI (This function is only supported on certain gimbals.)	55
2.1.15. SetTracking (This function is only supported on certain gimbals.)	56
2.1.16. GetDetection (This function is only supported on certain gimbals.)	58
2.2. AI Object Name Parameter Table	60
2.3. CGI Parameter-to-Value Table	61
3. Flight Controller Protocol Control Description	64
3.1. Communication Protocol Format	65
3.2. Message Data Definition	66
3.2.1. HEARTBEAT	66
3.2.2. ATTITUDE.....	67

3.2.3. GLOBAL_POSITION_INT.....	68
3.3. Checksum Calculation.....	70
3.3.1. C++ Example	70
3.3.2. Python Example	71
4. SBUS Protocol Control Description	73
4.1. Communication Protocol Format	74
4.2. Communication Protocol Example.....	75

1. Gimbal Control Description

This protocol is intended to define the communication protocol between a Personal Computer (PC) and the Gimbal. The relevant settings and precautions are as follows :

- **Data Format** : Little-Endian.
- **Network Transmission** : TCP / UDP packets, Hand-Shaking required between both parties if using TCP.
- **Network Address**: IP address is 192.168.144.200.
- **Network Port**: Both TCP and Using are using Port 2000 to used for data exchange or control commands between the PC and the Gimbal.

1.1. Communication Protocol Format

Address	Data Name	Data Type	Description
0-1	Header[2]	Unsigned char	0xAB, 0xCD
2	Protocol Version	Unsigned char	Protocol version
3	From Device ID	Unsigned char	Refer to Section 1.2 Device ID Definition
4	To Device ID	Unsigned char	Refer to Section 1.2 Device ID Definition
5	Sequence	Unsigned char	Packet sequence number
6	Length	Unsigned char	Length of Data
7	Message ID	Unsigned char	Command ID
8~ (8+Length)	Data[Length]	Refer to Section 1.3 Message Data Definition	Refer to Section 1.3 Message Data Definition
(8+Length)+1	Checksum	Unsigned char	Sum of all bytes

1.1.1. Protocol Packet Examples

Commanding the payload to start video recording:

Example packet as follows:

0xAB	0xCD	0x01	0xA0	0x01	0x01	0x04	0x3A
Header		Protocol	From ID	To ID	Sequence	Length	Message ID
0x02	0x00	0x00	0x01	0x5C			
Data				Checksum			

Commanding the payload to follow the aircraft heading:

Example packet as follows:

0xAB	0xCD	0x01	0xA0	0x01	0xEA	0x01	0x31
Header		Protocol	From ID	To ID	Sequence	Length	Message ID
0x01	0x37						
Data	Checksum						

Receiving the current status from the gimbal:

Example packet as follows:

0xAB	0xCD	0x01	0x01	0xA0	0xBD	0x22	0xF0
Header		Protocol	From ID	To ID	Sequence	Length	Message ID
0x17	0xDE	0xFF	0xFF	0x8C	0x00	0x00	0x00
Data : Roll Angle				Data : Yaw Angle			
0x8E	0xD1	0xFF	0xFF	0x0A	0x00	0x0A	0x00
Data : Pitch Angle				Data : EO Zoom Ratio		Data : IR Zoom Ratio	
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Data : Target Distance				Data : Target Altitude			
0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00
Data : Target Longitude				Data : Target Latitude			
0x12	0x51	0x3C					
Data : Recv Hz	Data : Status	Checksum					

1.2. Device ID Definition

ID	Corresponding Device
0x01	Gimbal
0xA0	PC

1.3. Message Data Definition

1.3.1. Message Response

Value Definition: 0x01

Function Description: Returns the current message ID received from the client.

Application Scenario: Used to acknowledge and echo the message ID back to the client, allowing the client to confirm that the packet was successfully transmitted.

Transmitting Device: Gimbal

Receiving Device: PC

Transmission Timing: Sent as required

Address	Data Name	Data Type	Description
0	Message ID	Char	Refer to Section 1.3 Message Data Definition

1.3.2. Follow Random Head

Value Definition: 0x31

Function Description: Sets the gimbal heading to align with the aircraft nose.

Application Scenario:

When set to 1, the gimbal heading will always follow the aircraft nose.

When set to 0, the gimbal will not follow the aircraft nose.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required

Address	Data Name	Data Type	Description
0	Follow Mode	Char	0: Do not follow aircraft nose 1: Always follow aircraft nose

1.3.3. Control Gimbal Velocity

Value Definition: 0x32

Function Description: Controls the rotation of the gimbal.

Application Scenario: Used in velocity mode to control gimbal rotation.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required

Data Format:

Address	Data Name	Data Type	Description		
0 ~ 3	Yaw Axis Velocity	Float	Range: -100 ~ 100 degrees/sec		
4 ~ 7	Pitch Axis Velocity	Float	Range: -100 ~ 100 degrees/sec		
8	Setting Flags	Char	Bit 0	Follow aircraft nose	1 = enable
			Bit 1 ~ 7	Reserved	

1.3.4. Transfer to Specify Angle

Value Definition: 0x33

Function Description: Controls the rotation of the gimbal.

Application Scenario: Used in angle mode to control gimbal rotation.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required

Data Format:

Address	Data Name	Data Type	Description		
0 ~ 3	Yaw Axis Angle	Float	Unit : -180 ~ 180 degrees		
4 ~ 7	Pitch Axis Angle	Float	Unit : -90 ~ 90 degrees		
8	Setting Flags	Char	Bit 0	Enable yaw axis	1 = enable
			Bit 1	Enable pitch axis	1 = enable
			Bit 2	Follow aircraft nose	1 = enable
			Bit 3~7	Reserved	

1.3.5. Camera Control

Value Definition: 0x3A

Function Description: Controls the gimbal camera.

Application Scenario: Controls zoom, focus, and recording switch of the visible-light camera within the gimbal.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required

Data Format:

Address	Data Name	Data Type	Description
0	Mode Selection	Char	0 : Zoom control 1 : Focus control 2 : Recording switch
1	Zoom Control	Char	0 : Stop zoom 1 : Zoom in 2 : Zoom out
2	Focus Control	Char	0 : Stop focus 1 : Focus far 2 : Focus near 3 : Single auto focus
3	Recording Switch	Char	0 : Stop recording 1 : Start recording

1.3.6. Camera Capture

Value Definition: 0x3B

Function Description: Controls the gimbal camera to take a photo.

Application Scenario: Commands the gimbal camera to capture a single image.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required Data Format:

Address	Data Name	Data Type	Description
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1.3.7. Thermal Control

Value Definition: 0x3C

Function Description: Controls the gimbal thermal imaging camera.

Application Scenario: Controls zoom, focus, and palette adjustment of the thermal imaging camera within the gimbal.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as requiredData Format:

Address	Data Name	Data Type	Description
0	Mode Selection	Char	0: Zoom control 1: Focus control 2: Palette adjustment
1	Zoom Control	Char	0: Stop zoom 1: Zoom in 2: Zoom out
2	Focus Control	Char	0: Stop focus 1: Focus + 2: Focus -
3	Palette Adjustment	Char	1: White Hot 2: Black Hot 3: Rainbow 4: Iron Red 5: Lava

1.3.8. Distance Measurement

Value Definition: 0x50

Function Description: Controls the gimbal distance measurement function.

Application Scenario: Enables or disables the gimbal distance measurement function.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Sent as required

Data Format:

Address	Data Name	Data Type	Description
0	Distance Measurement Switch	Char	0: Disable distance measurement 1: Enable distance measurement

1.3.9. UAV Status Return

Value Definition: 0x60

Function Description: Transmits the current UAV status to the gimbal.

Application Scenario: When UART communication with the flight controller is unavailable, this command continuously sends UAV information such as GPS coordinates, heading angle, and other flight data to the gimbal.

Transmitting Device: PC

Receiving Device: Gimbal

Transmission Timing: Continuous transmission at 10 Hz

Data Format:

Address	Data Name	Data Type	Description
0 ~ 3	UAV Longitude	Float	Unit: 1e-7 degrees
4 ~ 7	UAV Latitude	Float	Unit: 1e-7 degrees
8 ~ 11	Altitude (MSL)	Float	Unit: meters
12 ~ 15	UAV Heading	Float	Unit: degrees, Range: 0 ~ 360
16 ~ 19	Horizontal Speed	Float	Unit: meters/second
20 ~ 23	Distance to Home	Int32	Unit: meters
24 ~ 27	Battery Voltage	Float	Unit: volts
28	Satellite Count	UInt8	Unit: count

29	Reserved	UInt8	Default: 1
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1.3.10. Get Control IP

Value Definition: 0xC0

Function Description: Retrieve the network location of the PTZ (pan-tilt-zoom) controller.

Application Scenario: Using the UDP protocol, send to 255.255.255.255:2000 to obtain the PTZ control IP address, subnet mask, default gateway, and related information. The response shall follow the format specified in Section 1.3.10.

Transmitting Device: PC

Receiving Device: PTZ Controller

Transmission Timing: Send as needed

Data Format:

Address	Data Name	Data Type	Description
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1.3.11. Return / Change Control IP

Value Definition: 0xC1

Function Description: Modify the network location of the PTZ (pan-tilt-zoom) controller.

Application Scenario: Modify the PTZ control IP address, subnet mask, default gateway, and related information. The response shall follow the format specified in Section 1.3.11.

Transmitting Device: PC

Receiving Device: PTZ Controller

Transmission Timing: Send as needed

Data Format:

Address	Data Name	Data Type	Description
0 ~ 3	IP Address	Char	Range : 0.0.0.0 ~ 255.255.255.255
4 ~ 7	Subnet Mask	Char	Range : 0.0.0.0 ~ 255.255.255.255
8 ~ 11	Default Gateway	Char	Range : 0.0.0.0 ~ 255.255.255.255

1.3.12. Change Control IP Response

Value Definition: 0xC2

Function Description: Indicate successful modification of the PTZ (pan-tilt-zoom) controller network location.

Application Scenario: Return confirmation of the successfully updated PTZ control IP address, subnet mask, default gateway, and related information.

Transmitting Device: PTZ Controller

Receiving Device: PC

Transmission Timing: Send as needed

Data Format:

Address	Data Name	Data Type	Description
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1.3.13. Gimbal Status Return

Value Definition: 0xF0

Function Description: Return the current status of the PTZ (pan-tilt-zoom) controller.

Application Scenario: Continuously report the current PTZ angles, zoom level, as well as IMU calibration and ranging results.

Transmitting Device: PTZ Controller

Receiving Device: PC

Transmission Timing: Continuous transmission (frequency: 10 Hz)

Data Format:

Address	Data Name	Data Type	Description
0 ~ 3	Roll Axis Angle	Float	Unit: degrees Range: -90 ~ 90°
4 ~ 7	Pitch Axis Angle	Float	Unit: degrees Range: -90 ~ 90°
8 ~ 11	Yaw Axis Angle	Float	Unit: degrees Range: -180 ~ 180°
12 ~ 13	Visible Light Zoom Level	Int16	Unit: 0.1×
14 ~ 15	Thermal Imaging Zoom Level	Int16	Unit: 0.1×

16 ~ 19	Target Distance	Float	Unit: meters		
20 ~ 23	Target Altitude	Float	Unit: meters		
24 ~ 27	Target Longitude	Int32	Unit: 1e-7 degrees		
28 ~ 31	Target Latitude	Int32	Unit: 1e-7 degrees		
32	Data Reception Frequency	Char	Unit: Hz		
33	Data Flag	Char	Bit 0	IMU Calibration	True if normal
			Bit 1	Pitch Axis Status	
			Bit 2	Roll Axis Status	
			Bit 3	Yaw Axis Status	
			Bit 4	PTZ Mode	0 : Free Mode 1 : Follow Aircraft Nose
			Bit 5	Ranging Result	0 : No Ranging 1 : Successful Ranged
			Bit 6 ~ 7	Reserved	

1.3.14. Gimbal Version Return

Value Definition: 0xF1

Function Description: Return the version of current gimbal.

Application Scenario: When Get ACK Command (0xF2), Return the firmware version, gimbal model, and gimbal SN number of current gimbal.

Transmitting Device: PTZ Controller.

Receiving Device: PC

Transmission Timing: Send as needed

Data Format:

Address	Data Name	Data Type	Description
0	Firmware Version	Unsigned char	Example: 0b0000_0001 First 4 Bits : Minor Version Last 4 Bits : Major Version
1 ~ 4	Firmware Patch	int	Example : 251001 25 : Build Year 10 : Build Month 01 : Build Date
5 ~ 20	Gimbal Model Type	Char[16]	Example: KTG_TT3002

21 ~ 22	Gimbal SN	Unsigned short	Range: 0 ~ 65535
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1.3.15. Get Gimbal Version

Value Definition: 0xF2

Function Description: Ack firmware version of current gimbal.

Application Scenario: To Get Gimbal Firmware Version, SN Number, And Gimbal Type.

Transmitting Device: PC

Receiving Device: PTZ Controller.

Transmission Timing: Send as needed

Data Format:

Address	Data Name	Data Type	Description
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1.4. Checksum Calculation

The checksum for the current gimbal control protocol is calculated using a simple summation method. Example implementations are as follows :

1.4.1. C++ Example

```
unsigned char crc = 0;
for (unsigned char i = 0; i < length; i++)
{
    crc += buffer[i];
}
return crc;
```

1.4.2. Python Example

```
def calculate_crc(buffer):
    crc = 0
    for b in buffer:
        crc = (crc + b) & 0xFF
    return crc
```

2. Camera Control Description

This protocol defines the communication between a Personal Computer (PC) and the camera within the gimbal. The relevant settings and precautions are as follows :

- Data Type: Plain text string.
- Network Transmission: CGI (sent via HTTP).
- Network Address: IP address 192.168.144.108.
- Network Port: Port 80, used for data exchange or control commands between the PC and the camera.
- Authentication: Not required if absent; otherwise, Basic Auth is used.

2.1. Message Data Definition

2.1.1. SetPTZ

Function Description: Control The visible camera zoom, focus, and focus mode.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Parameter	Value Range	Parameter Description
Zoom_Ratio	1 ~ 42	Visible Zoom Directly.
PTZSpeed	1 ~ 9	Zoom / Focus Speed. 1 is slow, 9 is fast.
AF_Mode	0 ~ 1	0: normal AF mode. 1: auto-focus after zoom.
FocusMode	0 ~ 2	0: Auto 1: Manual 2: Push AF
Dir	ZoomStop: Stop Continuous Zoom. ZoomIn: Continuous Zoom-In.	

	ZoomOut: Continuous Zoom-Out. FocusStop: Stop Continuous Focus. FocusFar: Continuous Focus-Far. FocusNear: Continuous Focus-Near.
--	---

Sent Message :

<http://192.168.144.108/SetPTZ.cgi?Dir=ZoomIn>

Returned Message :

200 is OK

2.1.2. SetThermal

Function Description: Control The thermal camera zoom and change thermal colorpalette.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Parameter	Value Range	Parameter Description
colorpalette	1 ~ 5	1: Rainbow 2: Whitehot 3: Blackhot 4: Ironbow 5: Lava
zoom_ratio	1 ~ 16	Thermal Zoom Directly

Sent Message :

http://192.168.144.108/SetThermal.cgi?zoom_ratio=2

Returned Message :

200 is OK

2.1.3. GetAdjust

Function Description: Retrieves the current image quality settings of the camera.

Application Scenario: Used to obtain settings such as brightness, contrast, saturation, hue, and other image parameters.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
Brightness	0~31	Image brightness
Contrast	0~31	Image contrast
Hue	0~31	Hue / phase
Saturation	0~31	Image saturation
Shutter	0~24	Shutter time (exposure time)
AntiFog	0 : Disable 1 : Enable	Anti-fog switch
AntiFog_level	0~12	Anti-fog level
i_BLC	0~3	Backlight compensation

Denoise_3D	0~9	3D digital noise reduction
Denoise_2D	0~9	2D digital noise reduction
AGC	6,8,10,12,14,16,20,24,32,40,48,56,64,72,80,88,96	Automatic gain control
DayNight_Day	0: Color mode 2: B/W mode	Day/Night switch
dis	0: Disable 1: Enable	Digital image stabilization (EIS)

Sent Message :

<http://192.168.144.108:80/GetAdjust.cgi>

Returned Message :

Brightness=16
Contrast=16
Hue=16
Saturation=16
Sharpness=0
Anti

Fog=0
i_BLC=0
Denoise_3D=0
Denoise_2D=0
DayNight_Day=0
dis=

0

2.1.4. SetAdjust

Function Description: Sets the current image quality settings of the camera.

Application Scenario: Configures camera parameters such as brightness, contrast, saturation, hue, and other image settings.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Data Format:

Parameter	Value Range	Parameter Description
Brightness	0~31	Image brightness
Contrast	0~31	Image contrast
Hue	0~31	Hue / phase
Saturation	0~31	Image saturation
Shutter	0~24	Shutter time (exposure time)
AntiFog	0 : Disable 1 : Enable	Anti-fog switch
AntiFog_level	0~12	Anti-fog level
i_BLC	0~3	Backlight compensation

Denoise_3D	0~9	3D digital noise reduction
Denoise_2D	0~9	2D digital noise reduction
AGC	6~96	Automatic gain control
DayNight_Day	0: Color mode 2: B/W mode	Day/Night switch
dis	0: Disable 1: Enable	Digital image stabilization (EIS)

Sent Message :

<http://192.168.144.108/SetAdjust.cgi?Brightness=18&Hue=14>

Returned Message :

200 is OK

2.1.5. GetNetwork

Function Description: Retrieves the current network configuration of the camera.

Application Scenario: Obtains camera settings such as IP address, subnet mask, default gateway, and other network parameters.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
Ipaddr	0.0.0.0 ~ 255.255.255.255	IP Address
Netmask	0.0.0.0 ~ 255.255.255.255	Subnet Mask
Gateway	0.0.0.0 ~ 255.255.255.255	Default Gateway
Dns1	0.0.0.0 ~ 255.255.255.255	Primary DNS
Dns2	0.0.0.0 ~ 255.255.255.255	Secondary DNS

Transmission Example :

<http://192.168.144.108:80/GetNetwork.cgi>

Returned Message :

Ipaddr=192.168.144.108
Netmask=255.255.255.0
Gateway=192.168.144.1
Dns
1=8.8.8.8
Dns2=168.95.1.1

2.1.6. SetNetwork

Function Description: Sets the current network configuration of the camera.

Application Scenario: Configures camera settings such as IP address, subnet mask, default gateway, and other network parameters.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Data Format:

Parameter	Value Range	Parameter Description
Ipaddr	0.0.0.0 ~ 255.255.255.255	IP Address
Netmask	0.0.0.0 ~ 255.255.255.255	Subnet Mask
Gateway	0.0.0.0 ~ 255.255.255.255	Default Gateway
Dns1	0.0.0.0 ~ 255.255.255.255	Primary DNS
Dns2	0.0.0.0 ~ 255.255.255.255	Secondary DNS

Transmission Example :

<http://192.168.144.108:80/SetNetwork.cgi?Ipaddr=192.168.2.60&Gateway=192.168.2.1>

Returned Message :

200 is OK

2.1.7. GetMultimedia

Function Description: Retrieves the current image quality settings of the camera.

Application Scenario: Obtains camera parameters such as resolution, frame rate, compression ratio, and other image settings.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
Resolution	3840x2160 2560x1440 1920x1080 1280x720 640x480	Image Resolution
Fps	5 ~ 30	Frame Rate
Bitrate	256 ~ 4000	Compression Ratio
Mode	h265 h264	Video/Streaming Format

Transmission Example :

<http://192.168.144.108:80/GetMultimedia.cgi?CH={CH}>

Parameter	Value Range	Parameter Description
CH	0: EO Main → EO Main (Visible-light Recording) 1: EO Sub 1 → EO Sub 1 (Visible-light Downlink 1) 2: EO Sub 2 → EO Sub 2 (Visible-light Downlink 2) 3: IR Main → IR Main (Thermal Imaging Recording & Downlink)	Video Channel

Returned Message :

Resolution=1920x1080
Fps=30
Bitrate=4000
Mode=h265

2.1.8. GetImage

Function Description: Get the current snapshot by the camera.

Application Scenario: Get the current camera snapshot (EO / IR · Hint : The function can't save the Image to SD card).

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Data Format:

Parameter	Value Range	Parameter Description
CH	0: EO Main → EO Main (Visible-light Recording) 1: EO Sub 1 → EO Sub 1 (Visible-light Downlink 1) 2: EO Sub 2 → EO Sub 2 (Visible-light Downlink 2) 3: IR Main → IR Main (Thermal Imaging Recording & Downlink)	Video Channel
Resolution	3840x2160 2560x1440 1920x1080 1280x720 640x480	Image Resolution (Hint : Only 640x480 & 640x360 If CH=3)

Transmission Example :

<http://192.168.144.108:80/SetMultimedia.cgi?CH=1&Fps=15&Bitrate=1024>

Returned Message :

The current snapshot of the camera.

2.1.9. SetMultimedia

Function Description: Sets the current image quality settings of the camera.

Application Scenario: Configures camera parameters such as resolution, frame rate, compression ratio, and other image settings.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Data Format:

Parameter	Value Range	Parameter Description
CH	0: EO Main → EO Main (Visible-light Recording) 1: EO Sub 1 → EO Sub 1 (Visible-light Downlink 1) 2: EO Sub 2 → EO Sub 2 (Visible-light Downlink 2) 3: IR Main → IR Main (Thermal Imaging Recording & Downlink)	Video Channel
Resolution	3840x2160 2560x1440 1920x1080 1280x720 640x480	Image Resolution
Fps	5 ~ 30	Frame Rate
Bitrate	256 ~ 4000	Compression Ratio
Mode	h265	Video/Streaming Format

	h264	
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Transmission Example1 :

<http://192.168.144.108:80/SetMultimedia.cgi?CH=1&Fps=15&Bitrate=1024>

Returned Message :

200 is OK

Transmission Example2 :

<http://192.168.144.108:80/SetMultimedia.cgi?CH=2&Mode=h264>

Returned Message :

200 is OK

2.1.10. GetPlayBack

Function Description: Retrieves the current recorded video or captured photo stored in the camera.

Application Scenario: Obtains specific video recordings or photo files stored in the camera.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Transmission Example 1:

<http://192.168.144.108:80/GetPlayBack.cgi?Time=20250617>

Parameter	Value Range	Parameter Description
Time	19970101 ~ 20380119	Date

Returned Message :

avi/20250617/TEST_123_Thermal_20250617144636.jpg

avi/20250617/TEST_123_Thermal_20250617144637.jpg

avi/20250617/TEST_123_Thermal_20250617144638.jpg

avi/20250617/TEST_123_Thermal_20250617144639.jpg

avi/20250617/TEST_123_Thermal_20250617144640.jpg

avi/20250617/TEST_123_Vision_20250617144636.jpg

avi/20250617/TEST_123_Vision_20250617144637.jpg

avi/20250617/TEST_123_Vision_20250617144638.jpg

avi/20250617/TEST_123_Vision_20250617144639.jpg

avi/20250617/TEST_123_Vision_20250617144640.jpg

avi/20250617/TEST_123_Thermal_20250617093623.mp4

avi/20250617/TEST_123_Thermal_20250617093755.mp4

avi/20250617/TEST_123_Thermal_20250617093927.mp4

avi/20250617/TEST_123_Vision_20250617093623.mp4

avi/20250617/TEST_123_Vision_20250617093755.mp4

avi/20250617/TEST_123_Vision_20250617093927.mp4

Transmission Example 2 :

http://192.168.144.108:80/GetPlayBack.cgi?List=20250617_090000r

Parameter	Value Range	Parameter Description
List	19970101 ~ 20380119 000000 ~ 235959	Date + Time

Returned Message :

avi/20250617/090000r/TEST_123_Thermal_20250617093623.mp4

avi/20250617/090000r/TEST_123_Thermal_20250617093755.mp4

avi/20250617/090000r/TEST_123_Thermal_20250617093927.mp4

avi/20250617/090000r/TEST_123_Vision_20250617093623.mp4

avi/20250617/090000r/TEST_123_Vision_20250617093755.mp4

avi/20250617/090000r/TEST_123_Vision_20250617093927.mp4

2.1.11. GetSRTList

Function Description: Retrieves the current SRT subtitle files used for video recording in the camera.

Application Scenario: Obtains SRT subtitle files containing information such as gimbal PTZ angles, GPS data, and other relevant metadata.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Transmission Example:

http://192.168.144.108:80/GetSRTList.cgi?List=20250617_090000r

Parameter	Value Range	Parameter Description
List	19970101 ~ 20380119 000000 ~ 235959	Date + Time

Returned Message :

avi/srt/20250617/090000r/TEST_123_Thermal_20250617093623.srt

avi/srt/20250617/090000r/TEST_123_Thermal_20250617093755.srt

avi/srt/20250617/090000r/TEST_123_Thermal_20250617093927.srt

avi/srt/20250617/090000r/TEST_123_Thermal_20250617094059.srt

2.1.12. GetTime

Function Description: Retrieves the current time from the camera.

Application Scenario: Obtains information from the internal clock of the camera.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
TimeZone	+720,+660,+600,+540,+480, +420,+360,+300,+240,+210, +180,+120,+60,-0,-60,-120, -180,-210,-240,-270,-300, -330,-345,-360,-390,-420, -480,-540,-570,-600,-660, -720,-780	Time zone (unit: hours × minutes, e.g., GMT+8 = -480)
UseNtp	0: Disable 1: Enable	Enable or disable NTP time synchronization
Ntp_Update	0~172800	NTP Update Time

		Format: Seconds
NtpIP	0.0.0.0~255.255.255.255	IP address of the NTP server
Localtime	0 ~ 4294967295.123	Date and Time to Be Changed (Unit: Unix standard timestamp)

Transmission Example :

<http://192.168.144.108:80/GetTime.cgi>

Returned Message :

TimeZone=-480
UseNtp=0
NtpIP=198.123.30.132
Localtime=1747846406

2.1.13. SetTime

Function Description: Sets the current time in the camera.

Application Scenario: Adjusts and configures the internal clock of the camera.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: Admin

Transmission Example :

<http://192.168.144.108:80/SetTime.cgi?TimeZone=480&UseNtp=0&Localtime=1747846406>

Parameter	Value Range	Parameter Description
TimeZone	+720,+660,+600,+540,+480, +420,+360,+300,+240,+210, +180,+120,+60,-0,-60,-120, -180,-210,-240,-270,-300, -330,-345,-360,-390,-420, -480,-540,-570,-600,-660, -720,-780	Time Zone (Unit: hours × minutes, e.g., GMT+8 = -480)
UseNtp	0: Disable 1: Enable	Use NTP for time synchronization

Ntp_Update	0~172800	NTP Update Time Format: Seconds
NtpIP	0.0.0.0~255.255.255.255	NTP Server IP Address
Localtime	0 ~ 4294967295.123	Date and Time to Be Changed (Unit: Unix standard timestamp)

Returned Message :

200 is OK

2.1.14. SetAI (This function is only supported on certain gimbals.)

Function Description: Configures AI recognition and tracking functions.

Application Scenario: Enables or disables AI recognition and tracking functions within the camera.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
Trigger	1: Enable 0: Disable	Enable/Disable AI Recognition

Transmission Example :

<http://192.168.144.108:80/SetAI.cgi?Trigger=1>

Returned Message :

200 is OK

2.1.15. SetTracking (This function is only supported on certain gimbals.)

Function Description: Sets the target object for the camera to track.

Application Scenario: Uses an ID code to specify the object the camera should track.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
ID	-1 : Stop tracking 1 ~ 65535 : Target ID to track	Target ID
enable	0 : Stop tracking	
target_x1	0~8191	Top-Left X Coordinate For Manual Tracking
target_y1	0~8191	Top-Left Y Coordinate Manual Tracking
target_x2	0~8191	Bottom-Right X Coordinate Manual Tracking
target_y2	0~8191	Bottom-Right Y Coordinate Manual Tracking

Transmission Example For AI Tracking :

<http://192.168.144.108:80/SetTracking.cgi?ID=1>

Returned Message :

200 is OK

Transmission Example For Start Manual Tracking :

http://192.168.144.108:80/SetTracking.cgi?target_x1=0&target_y1=0&target_x2=2000&target_y2=2000

Returned Message :

200 is OK

Transmission Example For Stop Manual Tracking :

<http://192.168.144.108:80/SetTracking.cgi?enable=0>

Returned Message :

200 is OK

2.1.16. GetDetection (This function is only supported on certain gimbals.)

Function Description: Retrieves the image position of the currently tracked object by the camera.

Application Scenario: Obtains the image position of the object currently being tracked by the camera, allowing bounding box operations on the ground station.

Transmitting Device: PC

Receiving Device: Camera

Transmission Timing: Sent as required, approximately 10 Hz

Transmission Permission: None

Data Format:

Parameter	Value Range	Parameter Description
ID	1 ~ 65535	Detection ID
Type	1 ~ 99	Object Class (see Section 2.2)
x1	0~8191	Top-Left X Coordinate
y1	0~8191	Top-Left Y Coordinate
x2	0~8191	Bottom-Right X Coordinate
y2	0~8191	Bottom-Right Y Coordinate

Note: Uses a normalized coordinate system with the top-left corner of the image as the origin. All coordinate values range from [0, 8191].

Transmission Example :

<http://192.168.1.50/GetDetection.cgi>

Returned Message :

ID=1,Type=1,x1=100,y1=100,x2=200,y2=200
ID=2,Type=1,x1=200,y1=200,x2=300,y2=300
ID=3,Type=2,x1=300,y1=300,x2=400,y2=400

2.2. AI Object Name Parameter Table

Code	Tracked Object
0	Person
1	Bicycle
2	Car
3	Motorcycle
7	Truck
99	Tracking (Tracking)

2.3. CGI Parameter-to-Value Table

Image Quality Parameters	Parameter Overview
Brightness	-4(0-3),-3(4-7),-2(8-11),-1(12-15), 0(16-19),1(20-23),2(24-27), 3(28-29),4(30-31)
Contrast	-4(0-3),-3(4-7),-2(8-11),-1(12-15), 0(16-19),1(20-23),2(24-27), 3(28-29),4(30-31)
Hue	-4(0-3),-3(4-7),-2(8-11),-1(12-15), 0(16-19),1(20-23),2(24-27), 3(28-29),4(30-31)
Saturation	-4(0-3),-3(4-7),-2(8-11),-1(12-15), 0(16-19),1(20-23),2(24-27), 3(28-29),4(30-31)
Shutter	Outdoor(0-0),Indoor(1-1),1/1(12-12),1/30(2-2),1/50(3-3),1/60(4-4),1/100(5-5),1/125(6-6),1/180(13-13),1/250(7-7),1/350(14-14),1/400(20-20),1/450(21-21),1/500(8-8),1/550(22-22),1/600(23-23),1/650(24-24),1/700(15-15),1/1000(9-9),1/2000(16-

	16),1/4000(17-17),1/5000(10-10),1/6000(18-18),1/8000(19-19),1/10000(11-11)
AntiFog	Enable(1-1),Disable(0-0)
AntiFog_level	0(60-60),1(70-70),2(80-80),3(90-90), 4(100-100),5(110-110),6(120-120),7(130-130), 8(140-140),9(150-150),10(160-160), 11(170-170),12(180-180)
i_BLC	Off(0-0),Weak(1-1),Middle(2-2),Strong(3-3)
Denoise_3D	Off(0-0),1(1-1),2(2-2),3(3-3),4(4-4),5(5-5),6(6-6),7(7-7),8(8-8),9(9-9)
Denoise_2D	Off(0-0),1(1-1),2(2-2),3(3-3),4(4-4),5(5-5),6(6-6),7(7-7),8(8-8),9(9-9)
AGC	6(6x),8(8x),10(10x),12(12x),14(14x),16(16x),20(20x),24(24x),32(32x), 40(40x),48(48x),56(56x),64(64x),72(72x),80(80x),88(88x),96(96x)
ModeSelector	0: Sunny 3: Cloudy 4: Haze
DayNight_Day	0: Color 2: Black & White

dis	0: Off 1: On
------------	-----------------

3. Flight Controller Protocol Control Description

This protocol defines the communication between the Flight Controller (FC) and the Gimbal. The relevant settings and precautions are as follows :

- **Data Type:** MAVLink 2 packet
- **Transmission Method:** UART (Universal Asynchronous Receiver Transmitter, asynchronous serial communication)
- **Transmission Rate:** 115200 bps
- **Transmission Settings:** 8 data bits, no parity, 1 stop bit

3.1. Communication Protocol Format

Address	Data Name	Data Type	Description
0	Header	Unsigned char	0xFD
1	Length	Unsigned char	Length of Data
2	Incompatibility Flags	Unsigned char	Always 0.
3	Compatibility Flags	Unsigned char	Always 0.
4	Sequence	Unsigned char	Packet Sequence Number
5	System ID	Unsigned char	System ID: Default is 1
6	Component ID	Unsigned char	Component ID: Default is 154
7~9	Message ID	Unsigned int	Message ID
10~ (10+Length)	Data[Length]	Refer to Section 3.2 Message Data Definition	Refer to Section 3.2 & 3.3 Message Data Definition
(10+Length)+1 ~ (10+Length)+2	Checksum	Unsigned short	CRC-16/MCRF4XX for the message (excluding Header), including CRC_EXTRA byte

3.2. Message Data Definition

3.2.1. HEARTBEAT

Value Definition: 0

Function Description: Provides the heartbeat information from the gimbal.

Application Scenario: This messages shows let flight controller knows gimbal exists.

Transmitting Device: Gimbal

Receiving Device: FC

Transmission Timing: Continuous transmission at 1 Hz

Data Format:

Address	Data Name	Data Type	Description
0	custom_mode	unsigned int	Reversed.
1	type	unsigned char	Vehicle or component type. For gimbal, its always 26.
2	autopilot	unsigned char	Autopilot type / class. For gimbal, its always 0.
3	base_mode	unsigned char	System mode bitmap. For gimbal, its always 0.
4	system_status	unsigned char	System status flag. For gimbal, its always 4.
5	mavlink_version	unsigned char	MAVLink version.

3.2.2. ATTITUDE

Value Definition: 30

Function Description: Provides the attitude information from the flight controller.

Application Scenario: Allows determination of the current aircraft attitude, including roll, pitch, yaw, and other related angles.

Transmitting Device: FC

Receiving Device: Gimbal

Transmission Timing: Continuous transmission at 20 Hz

Data Format:

Address	Data Name	Data Type	Description
0	time_boot_ms	unsigned int	System Time Since FC Startup (unit: ms)
1	roll	float	Aircraft Roll Angle (unit: rad)
2	pitch	float	Aircraft Pitch Angle (unit: rad)
3	yaw	float	Aircraft Yaw Angle (unit: rad)
4	rollspeed	float	Aircraft Roll Rate (unit: rad/s)
5	pitchspeed	float	Aircraft Pitch Rate (unit: rad/s)
6	yawspeed	float	Aircraft Yaw Rate (unit: rad/s)

3.2.3. GLOBAL_POSITION_INT

Value Definition: 33

Function Description: Provides GPS information from the flight controller.

Application Scenario: Allows determination of the current aircraft location, including GPS latitude, longitude, altitude above sea level, and other position information.

Transmitting Device: FC

Receiving Device: Gimbal

Transmission Timing: Continuous transmission at 10 Hz

Data Format:

Address	Data Name	Data Type	Description
0	time_boot_ms	unsigned int	System Time Since FC Startup (unit: ms)
1	lat	int	Latitude (unit: degE7)
2	lon	int	Longitude (unit: degE7)
3	lat	int	Altitude Above Sea Level (unit: mm)
4	relative_alt	int	Altitude Relative to Takeoff Point (unit: mm)
5	vx	short	Velocity Along Latitude Axis (North Positive, unit: cm/s)
6	vy	short	Velocity Along Longitude Axis (East Positive, unit: cm/s)

7	vz	short	Velocity Along Altitude Axis (Down Positive, unit: cm/s)
8	hdg	unsigned short	Vehicle Heading Angle (unit: degrees)

3.3. Checksum Calculation

The checksum for this protocol is calculated as follows :

3.3.1. C++ Example

```
#define MAV_MSGID_HEARTBEAT            0
#define MAV_MSGID_ATTITUDE             30
#define MAV_MSGID_GLOBAL_POSITION_INT  33

#define MAV_MSGID_HEARTBEAT_EXTRA_CRC    50
#define MAV_MSGID_ATTITUDE_EXTRA_CRC     39
#define MAV_MSGID_GLOBAL_POSITION_INT_EXTRA_CRC  104

unsigned char GetExtraCRC(unsigned int msg_id)
{
    switch (msg_id)
    {
        case MAV_MSGID_HEARTBEAT:
            return MAV_MSGID_HEARTBEAT_EXTRA_CRC;
        case MAV_MSGID_ATTITUDE:
            return MAV_MSGID_ATTITUDE_EXTRA_CRC;
        case MAV_MSGID_GLOBAL_POSITION_INT:
            return MAV_MSGID_GLOBAL_POSITION_INT_EXTRA_CRC;
        default:
            return 0; // Unknown Message ID
    }
}

// Accumulate CRC (official MAVLink implementation)
void crc_accumulate(unsigned char data, unsigned short * crcAccum)
{
    unsigned char tmp = 0;

    tmp = data ^ (unsigned char)(*crcAccum & 0xFF);
    tmp ^= (tmp << 4);
    *crcAccum = (*crcAccum >> 8) ^ (tmp << 8) ^ (tmp << 3) ^ (tmp >> 4);
}
```

```

}

// Initialize CRC
void crc_init(unsigned short *crcAccum)
{
    *crcAccum = X25_INIT_CRC;
}

// Compute MAVLink V1 CRC (Header + Payload + CRC_EXTRA)
unsigned short MavCalcCRC(const unsigned char* buffer,unsigned char length, unsigned
char crc_extra)
{
    unsigned short crc = 0;
    crc_init(&crc);

    // Process message excluding magic byte (start from buffer[1])
    for (unsigned short i = 1; i < length; i++)
    {
        crc_accumulate(buffer[i], &crc);
    }

    // Append CRC_EXTRA (message-specific byte)
    crc_accumulate(crc_extra, &crc);

    return crc;
}

```

3.3.2. Python Example

```

# Message IDs
MAV_MSGID_HEARTBEAT = 0
MAV_MSGID_ATTITUDE = 30
MAV_MSGID_GLOBAL_POSITION_INT = 33

# CRC_EXTRA values

```

```

MAV_MSGID_HEARTBEAT_EXTRA_CRC = 50
MAV_MSGID_ATTITUDE_EXTRA_CRC = 39
MAV_MSGID_GLOBAL_POSITION_INT_EXTRA_CRC = 104

def get_extra_crc(msg_id):
    if msg_id == MAV_MSGID_HEARTBEAT:
        return MAV_MSGID_HEARTBEAT_EXTRA_CRC
    elif msg_id == MAV_MSGID_ATTITUDE:
        return MAV_MSGID_ATTITUDE_EXTRA_CRC
    elif msg_id == MAV_MSGID_GLOBAL_POSITION_INT:
        return MAV_MSGID_GLOBAL_POSITION_INT_EXTRA_CRC
    return 0

def crc_accumulate(data, crcAccum):
    tmp = data ^ (crcAccum & 0xFF)
    tmp ^= (tmp << 4) & 0xFF
    crcAccum = ((crcAccum >> 8) ^
                ((tmp << 8) & 0xFFFF) ^
                ((tmp << 3) & 0xFFFF) ^
                ((tmp >> 4) & 0xFFFF))
    return crcAccum & 0xFFFF

def mavlink_crc(buffer, crc_extra):
    crc = X25_INIT_CRC
    # Skip First Byte (Heading / magic)
    for b in buffer[1:]:
        # 確保 b 為 0..255
        crc = crc_accumulate(b & 0xFF, crc)
    # Append CRC_EXTRA
    crc = crc_accumulate(crc_extra & 0xFF, crc)
    return crc

```

4. SBUS Protocol Control Description

This protocol defines the communication between the RC Controller (RC) and the Gimbal.

The relevant settings and precautions are as follows :

- **Data Type:** SBUS packet
- **Transmission Method:** UART (Universal Asynchronous Receiver Transmitter, asynchronous serial communication)
- **Transmission Rate:** 100000 bps
- **Transmission Settings:** 9 data bits, even parity, 2 stop bits
- **Transmission Frequency:** 100 Hz

4.1. Communication Protocol Format

Address	Data Name	Data Type	Description	
0	Header	Unsigned char	0x0F	
1~22	Channel Data	Unsigned char	16 channels, 11 bits per channel	
23	Flag	Unsigned char	7 ~ 4	Default value: 0x00
			3	Failsafe
			2	Frame_Lost
			1	Channel 17
			0	Channel 18
24	Footer	Unsigned char	0x00	

4.2. Communication Protocol Example

	7	6	5	4	3	2	1	0
0	Header (0x0F)							
1	Ch1							
2	Ch2					Ch1		
3	Ch3		Ch2					
4	Ch3							
...								
24	0	0	0	0	Failsafe	Frame_lost	Ch18	Ch17
25	Footer(0x00)							